



Intestinal and Tissue Nematodes

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. How does the life cycle (and mode of infection) of *Strongyloides stercoralis* differ from that of *Ascaris lumbricoides* and *Trichuris trichiura*?
2. Which two nematodes produce a pulmonary (larval migration) phase with eosinophilia, and what is this syndrome called?
3. What food exposure transmits *Trichinella spiralis*, and where do its larvae ultimately encyst?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

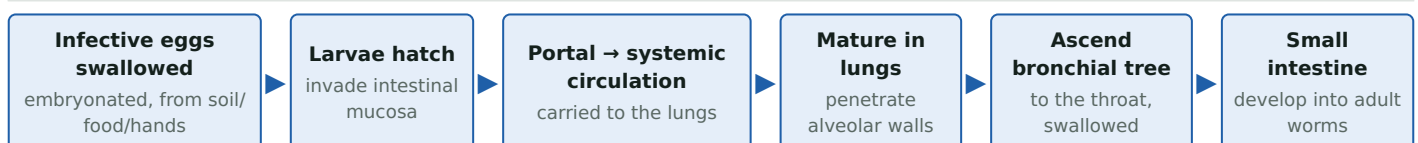
1 Nematode overview

Feature	Nematodes (roundworms)
Body plan	Unsegmented ; possess mouth, oesophagus, anus
Sexes	Generally separate sexes (unlike hermaphroditic trematodes/cestodes)
This lecture — intestinal/tissue nematodes covered	<i>Ascaris lumbricoides</i>, <i>Trichuris trichiura</i>, <i>Strongyloides stercoralis</i>, <i>Trichinella spiralis</i>

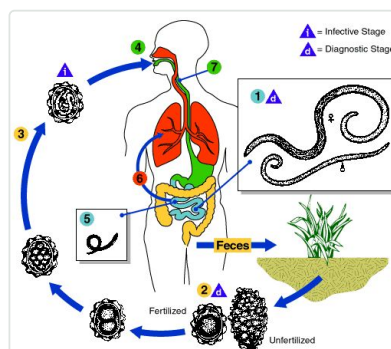
- Contrast: **trematodes** are unsegmented, leaf-like/cylindrical, generally hermaphroditic, eggs with lids, need ≥ 2 hosts (one a snail)
- **Cestodes** are segmented (scolex, neck, proglottids), hermaphroditic, have suckers/hooks, eggs generally have hooks and no lid

2 *Ascaris lumbricoides* — the giant roundworm

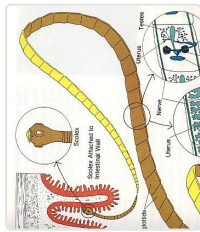
Feature	Detail
Size/host	Adults reach 40 cm ; live in small intestine lumen up to 2 years ; human is the definitive host, no intermediate host
Egg output	A female may produce ~200,000 eggs/day
Epidemiology	Most common human helminth infection ; worldwide, ~1 billion infected/year ; highest in tropical/subtropical, poor sanitation areas; more common in children (5-9 yrs) ; soil-transmitted



Unfertilized eggs are never infective; fertile eggs embryonate and become infective after several weeks in warm, moist soil.



Ascaris life cycle: soil-transmitted, human is the only host needed.



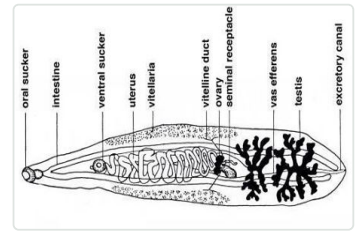
Labelled egg morphology.



Fertilized (rough-shelled) and unfertilized eggs on microscopy.



Mass of adult worms — gross specimen.



Abdominal X-ray: bowel loops packed with worms.

Phase	Presentation
Intestinal (adult worms)	Usually asymptomatic ; high worm burden → abdominal pain, intestinal obstruction, constipation ; may migrate → biliary tract obstruction, appendicitis, nasopharyngeal expulsion
Lung (larval migration)	Löffler syndrome : cough, dyspnea, hemoptysis, eosinophilic pneumonitis ; wheeze/bronchospasm (allergic); eosinophilia



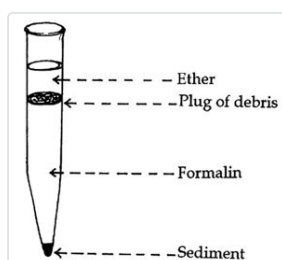
Massive Ascaris infection in child. A large bolus of roundworms expelled following antihelminthic treatment.

Chest X-ray during larval migration — pulmonary infiltrates of Löffler syndrome.

	Detail
Diagnosis	Microscopy — fertilized/unfertilized eggs or adults in feces; concentration method improves yield; larvae rarely in sputum
Treatment	Pyrantel pamoate (safe in pregnancy), mebendazole or albendazole
Prevention	Sanitary fecal disposal, food/hand hygiene, mass de-worming programmes

3 *Trichuris trichiura* — whipworm

Feature	Detail
Morphology	Adults ~ 4 cm , live in the colon ; anterior (whip-like) end threaded into the mucosa , thicker "handle" end free in the lumen
Egg output	3,000-20,000 eggs/day ; eggs passed non-embryonated , become infective in soil in 15-30 days
Transmission	Ingestion of soil-contaminated hands/food; eggs hatch in small intestine , larvae mature and settle as adults in the colon ; often co-infects with Ascaris (same soil-transmitted route)



Labelled adult morphology — female (long, whip-shaped) and male (coiled posterior end).



Classic lemon/barrel-shaped, bipolar-plugged egg.



Life cycle — soil-transmitted, ingestion of embryonated eggs.

Presentation	Detail
Mild/most infections	Most frequently asymptomatic
Heavy infestation (esp. children)	GI symptoms (tenderness, abdominal pain, chronic mucoid & bloody diarrhea), weight loss, weakness, growth retardation, iron-deficiency anemia , rectal prolapse

Detail

Diagnosis	Microscopy — lemon-shaped eggs or adult whipworms in feces (concentration method better); proctoscopy of rectal mucosa (or direct exam if prolapsed)
Treatment	Highly effective, 1-3 day course ; same dosing for adults and children

4 *Strongyloides stercoralis* — threadworm

- **Only adult females** live embedded in the small intestinal mucosa, reproducing by **parthenogenesis** (no male needed in the human host)
- **Complex life cycle** with free-living soil forms; **rhabditiform larvae** passed in feces → become **infective filariform larvae** either sexually (free-living cycle in soil) or **asexually within the intestine (autoinfection)**
- Filariform larvae reach the small intestine via **skin penetration** → **bloodstream**
- **Soil-transmitted**; worldwide prevalence **~100 million**



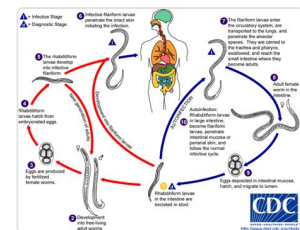
Life cycle: free-living and parasitic cycles, plus autoinfection loop.



Global distribution.



Larva currents — migrating serpiginous rash on trunk/buttocks.



Rhabditiform larva on stool microscopy.

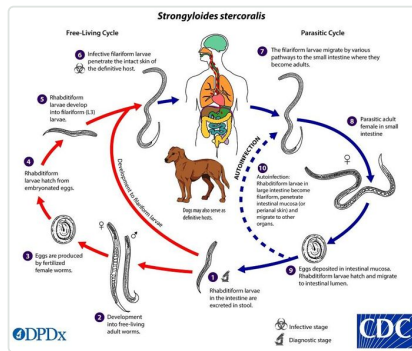
Presentation	Detail
Skin (entry)	Local irritation — rash, pruritus; larva currents on trunk/buttocks
Migrating larvae	Mild respiratory symptoms for a few weeks + asymptomatic eosinophilia (Löfller pneumonitis) ; rarely ectopic larvae in brain/other viscera
Established infection	May be asymptomatic
Severe infection	Epigastric pain, abdominal distension, vomiting, diarrhea with voluminous stools , malabsorption with dehydration/electrolyte disturbance, anemia

Hyperinfection syndrome — massive autoinfection in an **immunosuppressed host**: hemorrhagic diarrhea, intestinal perforation, hemorrhagic pneumonia, Gram-negative meningitis, shock, sepsis — **high mortality**. Up to 75% of chronic strongyloidiasis has mild eosinophilia or raised IgE.

	Detail
Diagnosis	Rhabditiform larvae in feces, or duodenal aspirate/sputum
Treatment	Mebendazole or thiabendazole

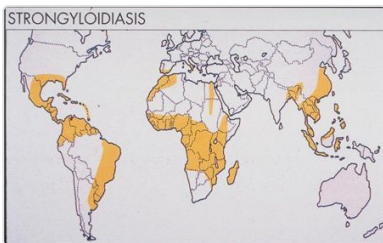
5 *Trichinella spiralis* — trichinellosis

- **Zoonotic**, worldwide; humans usually infected from **domestic pigs** (undercooked pork) — **viviparous** worm
- Larvae penetrate the intestinal wall → carried to **muscle cells**, where they **coil up** and become infective within **~3 weeks**
- New host ingests larvae in **undercooked meat** → released, develop into **adults in the intestine within about a day**; **adults live in the gut, larvae encyst in muscle**
- **Food-borne epidemics** can occur



Life cycle: pig → undercooked pork → human intestine → muscle encystment.

Phase	Presentation
Intestinal	Intestinal inflammation → diarrhea
Muscle invasion (larval migration, ~1 week)	Inflammatory response → periorbital oedema, subungual (splinter) haemorrhages , muscle pains, myocarditis ; especially muscles of respiration and tongue ; high eosinophilia ; conjunctivitis, myalgias
Long term	Fibrosis and degeneration of encysted larvae → calcification



Periorbital oedema — classic sign of larval migration.



Splinter (subungual) haemorrhages.



Encysted larvae within skeletal muscle (biopsy).

	Detail
Diagnosis	Clinical features + microscopic examination, serology, muscle biopsy
Treatment	Mebendazole or thiabendazole + steroids (for the inflammatory myositis)

P Must-remember pearls

- **Ascaris**: soil-transmitted, no intermediate host, 40 cm adults in small intestine; larval **lung migration** → **Löffler syndrome**; Rx **pyrantel pamoate/mebendazole/albendazole**.
- **Trichuris**: whipworm, colon, anterior end embedded in mucosa; lemon-shaped eggs; heavy infection → **bloody diarrhea, rectal prolapse, anemia**.
- **Strongyloides**: unique — **autoinfection** lets it persist for decades; **larva currens**; **hyperinfection syndrome** in immunosuppressed = high mortality; diagnose via **rhabditiform larvae** in stool (not eggs).
- **Trichinella**: zoonotic, from **undercooked pork**; larvae encyst in **muscle**; classic triad-type findings = **periorbital oedema + subungual haemorrhages + myositis/high eosinophilia**.
- Ascaris and Strongyloides both cause a **pulmonary larval migration phase with eosinophilia (Löffler-type)** — a shared exam trap between the two.

Q Test yourself

1. A child from an area with poor sanitation has abdominal pain and an abdominal X-ray shows bowel loops packed with worms up to 40 cm long. Which organism, and what is the key epidemiologic feature?

- A. *Ascaris lumbricoides* — no intermediate host needed, humans are the only host required
- B. *Trichuris trichiura* — requires a snail intermediate host
- C. *Strongyloides stercoralis* — requires ingestion of undercooked pork
- D. *Trichinella spiralis* — requires two definitive hosts

2. A patient with a helminth infection develops cough, dyspnea, hemoptysis, and eosinophilic pneumonitis several days after ingesting embryonated eggs. What is this syndrome called?

- A. Löffler syndrome
- B. Hyperinfection syndrome
- C. Halzoun
- D. Katayama fever

3. Which nematode is unique among those discussed for its ability to cause autoinfection and persist in a host for decades?

- | | |
|------------------------------|-------------------------|
| A. Strongyloides stercoralis | B. Ascaris lumbricoides |
| C. Trichuris trichiura | D. Trichinella spiralis |

4. A patient who recently ate undercooked pork develops periorbital oedema, splinter haemorrhages, and severe myalgia with marked eosinophilia. What is the diagnosis?

- | | |
|--|---------------------|
| A. Trichinellosis (Trichinella spiralis) | B. Ascariasis |
| C. Trichuriasis | D. Strongyloidiasis |

5. Which diagnostic finding is expected in stool microscopy for Strongyloides stercoralis, in contrast to the other nematodes discussed?

- | | |
|--|---|
| A. Rhabditiform larvae, not eggs | B. Lemon-shaped eggs with bipolar plugs |
| C. Fertilized/unfertilized eggs with a rough mammillated shell | D. Encysted larvae in muscle tissue |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — Ascaris is soil-transmitted with humans as the sole host; no intermediate host is needed.
- 2 — A** — Löffler syndrome refers to the pulmonary eosinophilic reaction during larval lung migration, classically seen with Ascaris.
- 3 — A** — Filariform larvae can form within the intestine itself (autoinfection), allowing Strongyloides to persist without re-exposure.
- 4 — A** — Periorbital oedema, subungual haemorrhages and myositis after undercooked pork ingestion are classic for Trichinella muscle-larval migration.
- 5 — A** — Only adult female Strongyloides live in the mucosa and release larvae, not eggs, into the stool.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Microbiology GIT Block · MEDucation by Mattin Majid · GIT SGL 8 slides, 2026-2027
Study alongside the lecture, not instead of it.